

CHyLL v2: Continuous Latent Embeddings of Hybrid Systems

What the model learns, and what each loss term enforces

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Hybrid Dynamical Systems

A *hybrid* dynamical system mixes smooth motion with sudden jumps.

- A walking robot: the swing leg moves smoothly, then strikes the ground and the state jumps.
- A bouncing ball: free flight under gravity, then an instantaneous change of velocity at each impact.

The jump is a genuine discontinuity. A smooth model of the dynamics cannot be fitted directly across it.

Three Example Systems

System	State dim.	Behavior
Rimless wheel	2	stable rolling limit cycle
Bouncing ball	2	bounces, losing energy at each impact
Compass-gait biped	4	period-doubling route to chaos

They increase in difficulty. The compass-gait biped walks with period 1, then 2, 4, 8, and finally chaotically as the ground slope steepens.

The Idea

Replace the discontinuous system with a continuous one that is easy to model.

- An **encoder** maps each state into a latent space.
- A single **smooth vector field** advances the latent state in time.
- A **decoder** maps the latent state back.

The jump is absorbed into the smooth latent flow. This follows Teng, Liu, and Sreenath (ICML 2026).

The Mapping-Cylinder State Space

To make the jump continuous, the state is augmented with one extra coordinate $s \in [0, 1]$:

$$y = (x, s) \in X'.$$

- $s = 0$: ordinary motion of the original system.
- s sweeping $0 \rightarrow 1$: the jump, unrolled as a continuous path.

On X' the trajectory is continuous, so a smooth latent model can be fitted to it.

The Learned Maps

CHyLL v2 learns three maps. Write $Z = \mathbb{R}^d$ for the latent space.

$E_\theta : X' \rightarrow Z$	encoder,
$V_\theta : Z \rightarrow Z$	latent vector field,
$D_\theta : Z \rightarrow X'$	decoder.

The time- τ flow of V_θ is written F_θ . The latent dimension is $d = 7$ for the rimless wheel and bouncing ball, and $d = 11$ for the compass-gait biped.

Training: The Forward Pass

For a sampled trajectory slice $y_0, \dots, y_{T-1}$:

$$z_i = E_\theta(y_i),$$

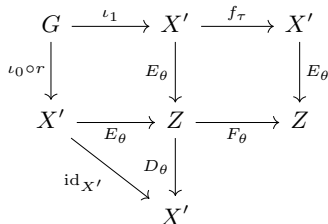
$$\hat{z}_i = F_\theta^i(z_0),$$

$$\hat{y}_i = D_\theta(\hat{z}_i).$$

Each state is encoded, the latent flow is integrated forward, and the integrated latent trajectory is decoded. The loss compares the encoded z_i , the integrated $\hat{z}_i$, and the decoded $\hat{y}_i$ against the data.

The Commutative Diagram

Here f_τ is one time- τ step of the system, G the impact set, r the reset map, and ι_0, ι_1 the inclusions at $s = 0$ and $s = 1$. The learned maps should make this diagram commute:



Each cell is enforced by one loss term:

- Left square, L_g : E_θ sends both sides of a reset to one latent point.
- Top-right square, L_z : the latent flow matches the dynamics.
- Lower triangle, L_x : the decoder inverts the encoder.

Two Conditions Beyond the Diagram

Commuting is not the whole story. Two more terms act at the seam where s wraps from 1 back to 0:

- L_v : the latent velocity has no kink crossing a seam.
- L_c : the encoder keeps all d latent coordinates active and does not collapse them.

These seams are detected from the sampled coordinate s in the data, not from symbolic guard equations.

What Each Loss Term Enforces

Term	Enforces	Code helper
L_x	decoder recovers the state	<code>reconstruction_loss</code>
L_z	encode/flow square commutes	<code>latent_consistency_loss</code>
L_g	reset pairs share a latent point	<code>gluing_loss</code>
L_v	latent velocity continuous at seams	<code>seam_velocity_loss</code>
L_c	latent coordinates do not collapse	<code>anti_collapse_loss</code>

L_g , L_z , and L_x are the three commuting cells of the diagram. L_v and L_c are the two conditions beyond it.

The Composite Objective

$$L = w_x L_x + w_z L_z + w_g L_g + w_v L_v + w_c L_c.$$

Setting	$(w_x, w_z, w_g, w_v, w_c)$
Rimless wheel, bouncing ball	(1, 1, 3, 1, 1)
Compass-gait biped	(1, 1, 3, 0, 1)

The exact values for any run are stored in its `config.json`.

Results

- **Rimless wheel.** The model reproduces the rolling limit cycle, and the learned embedding recovers its expected single topological cycle.
- **Compass-gait biped.** The cascade analysis recovers the period-doubling sequence exactly: the learned latent states form 2, 4, and 8 clusters at the period-2, -4, and -8 gaits, and disperse under chaos.
- **Bouncing ball.** The orbit spirals slowly toward rest rather than closing into a loop, so its topological signature is only partially recovered.

Where the Code Lives

Object	File
$E_\theta, V_\theta, D_\theta$	<code>networks.py</code>
latent ODE rollout	<code>ode.py</code>
loss terms L_x, L_z, L_g, L_v, L_c	<code>losses.py</code>
training loop	<code>train.py</code>
system simulators and reset maps	<code>systems/</code>

These files are the CHyLL v2 core library, under `chyll_v2/chyll_v2/`. Topology postprocessing lives under `chyll_v2/cycling_signature/`.

Summary

- A hybrid system is made continuous on the mapping cylinder X' .
- CHyLL v2 learns an encoder, a latent vector field, and a decoder.
- L_g , L_z , and L_x are the commuting cells of the diagram.
- L_v and L_c add the seam-regularity conditions.
- The learned embeddings recover the expected topology for the rimless wheel and the compass-gait cascade.